DAY — 09 SEAT NUMBER

2011	III	09	1100	J - 570	(E)

PHYSICS PAPER - II (54)

(Light, Electricity, Magnetism and Modern Physics)

Time : 2 Hrs. (4 Pages) Max. Marks : 40

Note : (i) Attempt all questions.

(ii) Neat diagrams must be drawn, wherever necessary.

(iii) Figures to the right indicate full marks.

(iv) Use of only logarithmic table is allowed.

(v) All symbols have their usual meanings unless otherwise stated.

(vi) Answer to every new question must be written on a new page.

Q. 1. Select and write the most appropriate answer from the given alternatives for each sub-question : [8]

(i) The apparatus used to produce plane polarised light is (1)

(a) nicol prism

(b) glass prism

(c) biprism

(d) hollow prism

(ii) A current in A. C. circuit lags behind the applied e. m. f. by 90°. The circuit contains (1)

(a) only resistance

(b) only capacitance

(c) resistance and capacitance

(d) only inductance

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(iii) Limit of resolution of an optical instrument is equal to (1)

(a) resolving power

(b) reciprocal of resolving power

(c) $\sqrt{\text{resolving power}}$

(d) $\frac{1}{\sqrt{\text{resolving power}}}$

(iv) The height of a geostationary satellite above the earth's surface is about (1)

(a) 36000 km. (b) 3600 km.

(c) 360 km. (d) 36 km.

(v) Condenser is a device used to store (1)

(a) large potential at low charge.

(b) low potential at low charge.

(c) large charge at low potential.

(d) large charge at large potential.

(vi) To convert galvanometer into voltmeter (1)

(a) large resistance should be connected in series.

(b) large resistance should be connected in parallel.

(c) low resistance should be connected in series.

(d) low resistance should be connected in parallel.

(vii) An electron moves through a crossed field of 5000 v/m and 5×10^{-4} wb/m^2, then its velocity is (1)

(a) 10^6 m/s (b) 10^7 m/s

(c) 10^8 m/s (d) 10^9 m/s

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(viii) In a potentiometer experiment balancing lengths for first cell and second cell are 2·5 m and 2 m respectively, when cells are used separately. Hence E_1/E_2 is (1)

(a) 0·8 (b) 1·0

(c) 1·1 (d) 1·25

Q. 2. (A) Attempt any ONE : [8]

(i) A capacitor carries a charge of 8 μc at a potential 400 V. How much electrostatic energy is stored in the capacitor? (2)

(ii) A series LCR circuit has capacitor of 0·2 μf and inductor of 8 mH. Find its resonant frequency. (2)

(B) Attempt any TWO :

(i) Describe Wheatstone's network with neat circuit diagram. Obtain expression for its balance condition. (3)

(ii) Describe J. J. Thomson's experiment to determine velocity of electron. (3)

(iii) Draw a neat circuit diagram for NPN transistor as common emitter amplifier and explain its working. (3)

Q. 3. (A) Attempt any ONE : [8]

(i) State and explain principle of potentiometer. (2)

(ii) Draw neat labelled ray diagrams to get magnified and diminished images of two virtual sources using convex lens in biprism experiment. (2)

(B) Attempt any TWO :

(i) State and prove Gauss' theorem in electrostatics. (3)

(ii) State different types of wave propagation. Describe sky wave propagation in brief. (3)

(iii) State the principle of moving coil galvanometer and explain its working. (3)

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Q. 4. (A) Attempt any TWO : [8]

(i) Show that the radius of Bohr orbit is directly proportional to the square of the principle quantum number. (2)

(ii) Draw a neat labelled circuit diagram of full wave rectifier using semiconductor diode. (2)

(iii) Show that accuracy of tangent galvanometer is maximum at deflection 45°. (2)

(B) Attempt any ONE :

(i) Derive the expression for magnitude and direction of magnetic induction at any point due to short magnetic dipole. (4)

(ii) Prove theoretically, the relation between e. m. f. induced and rate of change of magnetic flux in the electric circuit in electromagnetic induction. (4)

Q. 5. Attempt any TWO : [8]

(i) Monochromatic light from a narrow slit illuminates two narrow slits 0·3 mm apart producing an interference pattern with bright fringes 1·5 mm apart on a screen 75 cm away. Find the wavelength of light. How will the fringe width be altered if
(a) the distance of the screen is doubled.
(b) the separation between the slits is doubled. (4)

(ii) The wavelength of H_α line of Balmer series of Hydrogen spectrum is 6563 Å. Find the
(a) Wavelength of H_γ line of Balmer series,
(b) Shortest wavelength of Brackett series. (4)

(iii) The wavelength of a beam of light in air is 3750 Å. Find the number of waves of the beam in 10 cm of glass. Also find the time required by the beam to pass through 10 cm of glass of refractive index 1·5. (Velocity of light in air = 3×10^8 m/s) (4)

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